Mesa/Boogie Solo Control TM

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In 2004, Mesa/Boogie was awarded a patent for its Solo ControlTM technology to create a guitar amplifier with footswitchable volume boost:

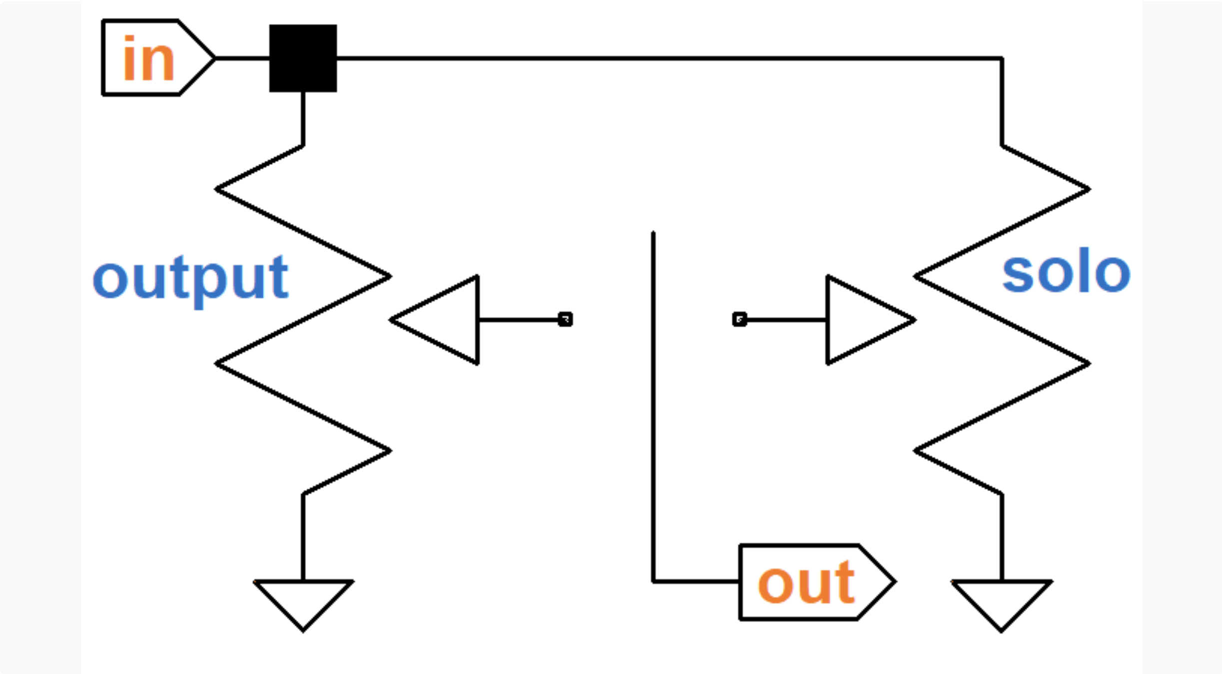
U.S. Patent 6,724,897

According to the patent, Randall Smith developed the technology in response to changes in musical styles. Traditionally, guitar amplifiers have a relatively clean channel ("rhythm mode") and a more distorted lead channel ("lead enhancement mode"). Contemporary styles incorporate amplifier distortion to various degrees for both rhythm and lead. This creates a need for footswitchable volume boost without affecting distortion.

"A guitarist may wish to 'step forward' in the band's mix for a spotlight feature in any mode, including modes used primarily for background playing. Thus, the ability to boost the volume level without switching to a different mode provides a unique and valuable capability."

A Simple Switch

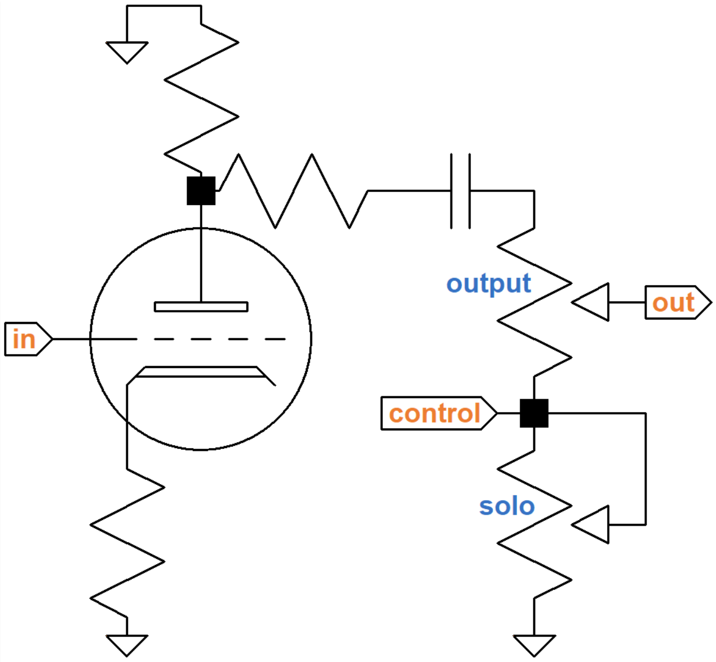
The simplest implementation is to switch between two volume controls: one sets the normal output level and the other sets a louder solo level.



A downside to this approach is that the user can just as easily set the solo level lower than the normal output level. The controls suffer from an identity crisis - labeling either of them as "solo" is completely arbitrary. The design can be tweaked, but there are better options.

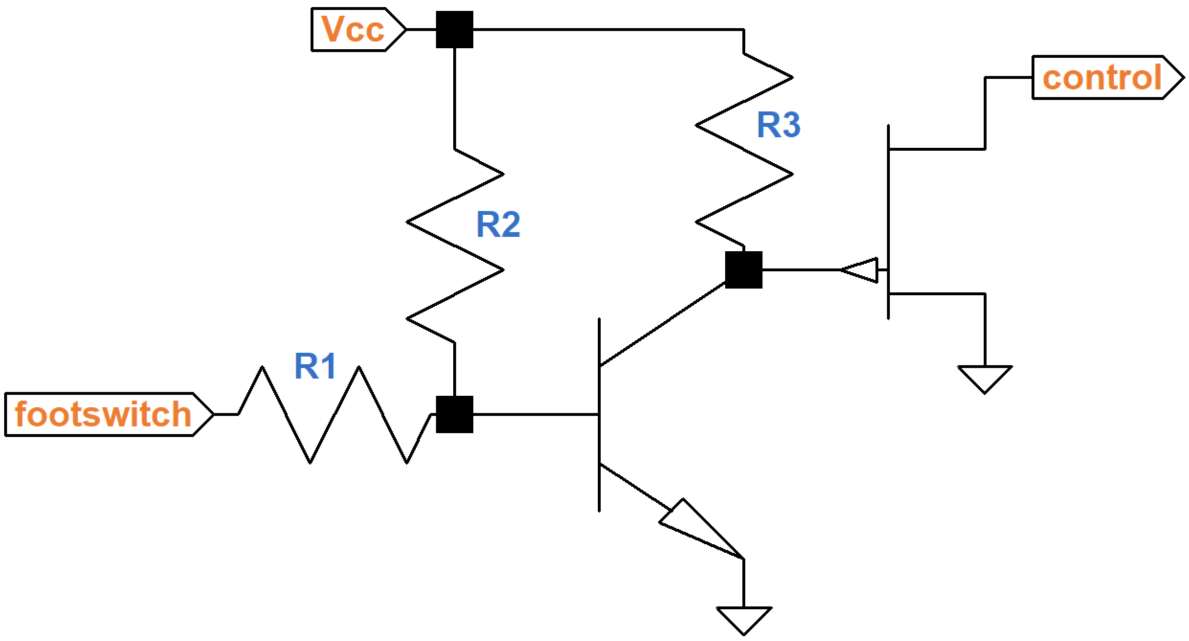
A Better Approach

The Mesa/Boogie MK V, Lonestar Special, and Dual Rectifier have a totem-pole architecture.



When `control` is grounded the solo control is shorted. When left open, the solo potentiometer adds variable resistance below the output control wiper for user-controlled volume boost.

The Road King II and other Mesa/Boogie models use semiconductor switching.



R2 has a resistance that is orders of magnitude greater than R1. They form a voltage divider for the base of the NPN transistor. When the footswitch is shorted to ground, the voltage at the base of the NPN transistor switches from nearly VCC to nearly 0V, turning the transistor off. This turns the p-channel JFET off, removing its short circuit across the solo potentiometer.